◆ ABU DHABI · 1948 · 2026 ◆

PRESENTED TO ADEK

A HERITAGE-ROOTED LEARNING PROTOTYPE

The Pearl *and* the Loom

اللؤلؤة والنَّول

A family story, woven row by row.

FROM

Bilal Saeed

42 ABU DHABI · LOGIN · BSAEED

FOR

ADEK

DEPARTMENT OF EDUCATION & KNOWLEDGE

Why I built *this, for you.*

ADEK's mandate is plain — raise outcomes for every learner in the Emirate without flattening what makes them *Emirati*. Most ed-tech I have used as a student does the first by sacrificing the second.

The Pearl and the Loom is a working prototype that refuses the trade-off. It teaches Grade 4 math and Grade 8 science through two threads of UAE heritage — Sadu weaving and the pearling season — in a single bilingual app.

THE CANDIDATE

Bilal Saeed

42 Abu Dhabi · Login Bsaeed

THE ROLE

Educational Technology Specialist

ADEK · Abu Dhabi

Imported curricula. Imported metaphors.

01

THE SCREEN

Most classroom apps in the Emirate were authored elsewhere. Animals are foxes and squirrels. Settings are forests and farmhouses. Numerals are Latin only.

02

THE COST

Children master the math but learn it as something *happening to other people, somewhere else.* Heritage gets relegated to social-studies hour.

03

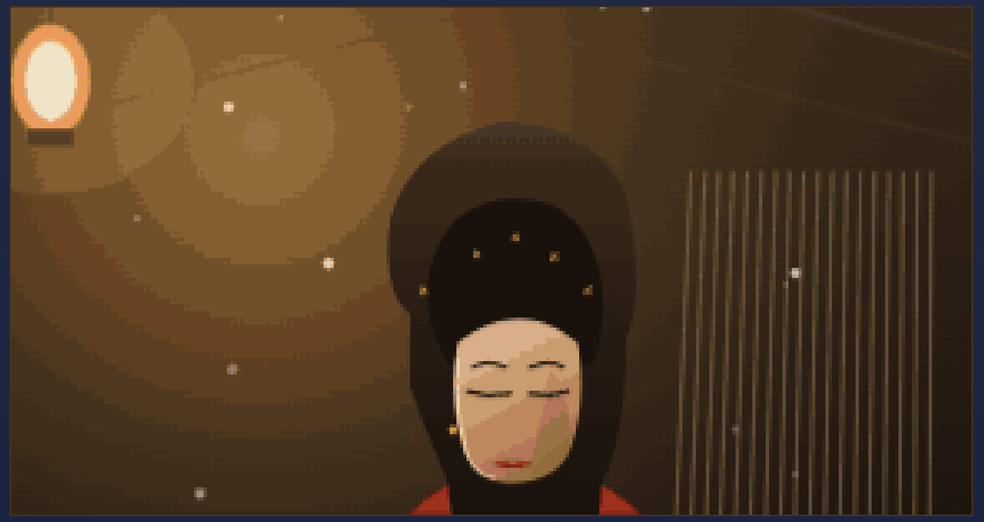
THE OPPORTUNITY

UAE heritage is, structurally, already a curriculum — geometry lives in a Sadu band; pressure and buoyancy live in a pearl diver's lungs.

Teach the curriculum through the heritage — not *around* it.

A Sadu loom is a coordinate plane. A dhow at season is a physics lab. We give two children — *Layla* and *Saif* — and their grandmother's unfinished heirloom as the spine. Every lesson earns one row, or one pearl.

WHOSE JOURNEY TODAY?



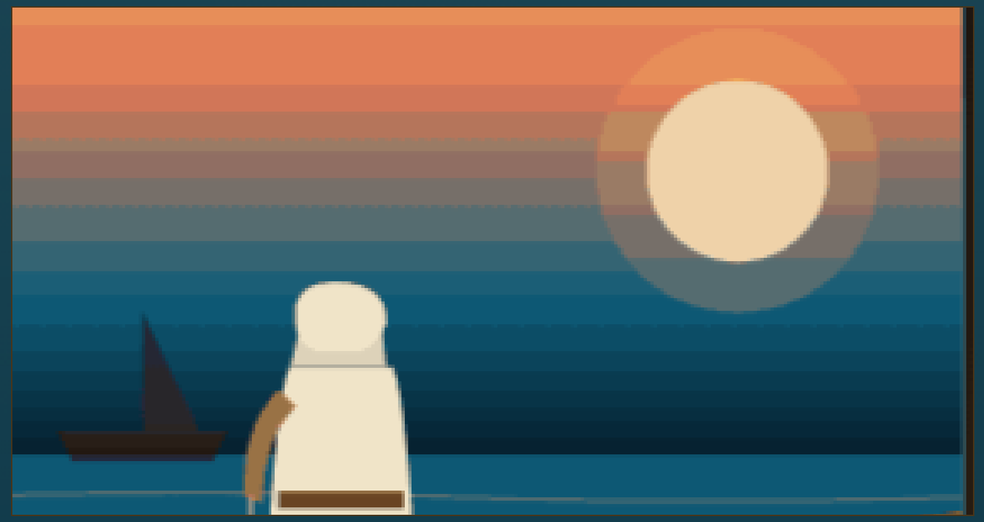
AT THE LOOM · GRADE 4 MATH

Layla

ليلى

Geometry, symmetry, fractions – taught as *rows of weave*. Each completed lesson is one band of her tapestry.

3/30
ROWS WOVEN



AT SEA · GRADE 8 SCIENCE

Saif

سيف

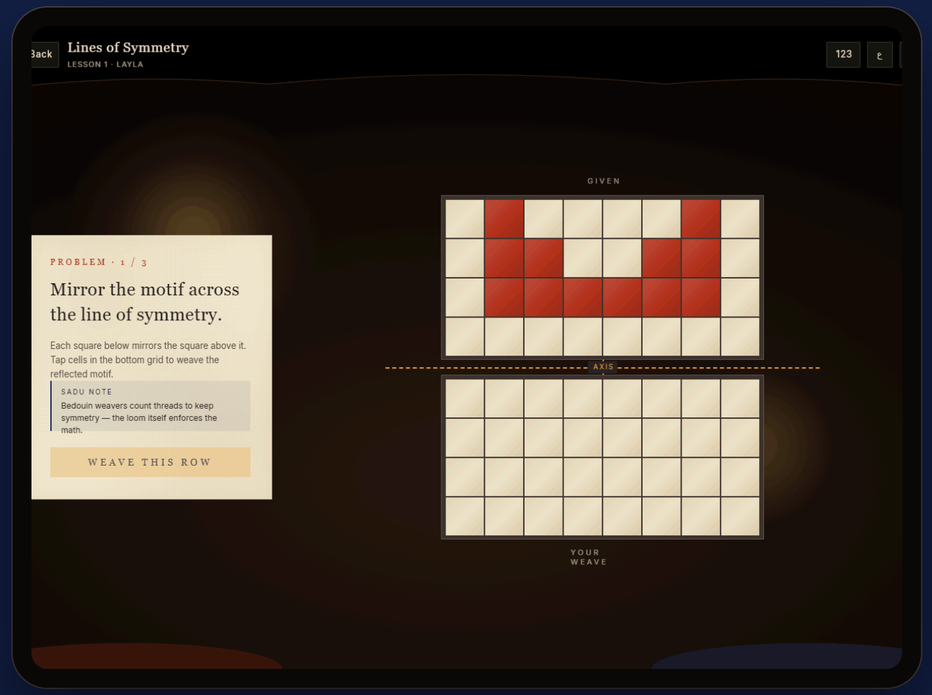
Pressure, buoyancy, marine life – taught through the *ghasa*, the dive. Each lesson surfaces with a pearl.

7/12
PEARLS COLLECTED

The loom is the math.

Sadu's geometry is exact: warp and weft cross at right angles, motifs are mirrored or repeated, bands divide into halves and quarters. That structure is the lesson plan.

LESSON	MATH OBJECTIVE	HERITAGE SURFACE
Symmetry	Identify lines of reflection; complete a mirrored figure.	Fold the motif. Layla's diamond is half-drawn — finish it so the warp lines up.
Fractions	Halves, thirds, quarters of a whole length.	Divide a band of weave into equal parts of color before the warp is set.
Tessellation	Tile a plane with repeating shapes; recognise translation & rotation.	Choose a unit motif. Watch it march across a finished band.
Coordinates · next	Plot points on a 2D grid.	The loom itself: warp = x, weft rows = y.



LESSON · IN APP

The dive is the science.

A pearl diver descending ten metres without a tank is a physics problem with stakes. Saif lives that problem each season — pressure, buoyancy, oxygen, ecosystem — and earns pearls for the family chest.

LESSON	SCIENCE OBJECTIVE	HERITAGE SURFACE
Pressure	Fluid pressure increases with depth; quantify in metres & atmospheres.	Saif descends. The hull, the rope, the lungs — read the gauge.
Buoyancy	Density & displacement; why a dhow floats and a stone does not.	Tie the diving stone. Calculate sink rate against current.
Oxygen · next	Breath-hold physiology; CO_2 tolerance.	Hold breath with the diver. The bar drains in real time.
Ecosystem · next	Reef food-web; species & their roles.	Identify what swims past on the way down.

THE DIVE · IN APP

One heirloom. Two children.

Layla and Saif share *one* family tapestry — a real Sadu hanging their grandmother began and never finished. Layla weaves the rows. Saif's pearls are braided into the rows she weaves.

Neither child can finish the heirloom alone. Math without science leaves blank spaces; science without math leaves loose threads. The shared object is the cooperation mechanic.

LAYLA WEAVES · SAIF BRAIDS IN

English & Arabic. Both, equally.

Not a translation layer bolted onto an English app — full RTL layout, Arabic-Indic numeral toggle, and Arabic typography (Tajawal) used at the same weight as the English face.

LAYOUT · LTR

Whose journey today?

Layla · *at the loom*

3 / 30 rows woven

LAYOUT · RTL

رحلة من اليوم؟

ليلى · *عند النَّول*

٣ / ٣٠ صفًّا منسوجًا

NUMERALS · TOGGLE

0–9 ٩-٠

One toggle in the corner. The math is the same; *the symbols belong to the child.*

Built for the iPad on the desk.

ADEK classrooms run on iPads. The prototype is fluid from iPad portrait through 1440 web, designed thumb-first, and bears no chrome a substitute teacher would have to learn.

IPAD · PORTRAIT

1024 × 1366

One-hand reading. Tapestry hangs full-length, like a real Sadu wall hanging.

IPAD · LANDSCAPE

1366 × 1024

The default classroom view. Lessons on one half, heritage canvas on the other.

WEB · 1440

1440 × 900

For administrators & teacher review. Same content, no compromise on density.

15 screens. One coherent prototype.

15

SCREENS SHIPPED

Home, two hubs, three lessons, four dive states, chest, braiding, styleguide.

2

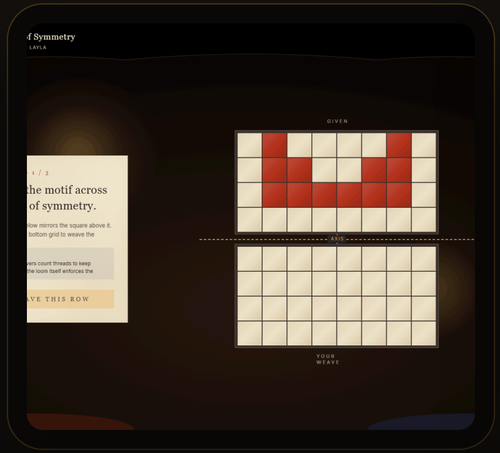
LANGUAGES · FULL RTL

English & Arabic, with Arabic-Indic numerals.

7

CURRICULUM OBJECTIVES

Mapped to MoE Grade 4 math & Grade 8 science strands.



From prototype to a pilot in one Abu Dhabi school.

PHASE 01 · Q3 2026

Curriculum lock

Map every screen to MoE outcomes with an ADEK subject lead. Lock copy in EN & AR with a teacher review panel.

PHASE 02 · Q4 2026

Heritage council

Sit with Sadu weavers & pearling families. Replace placeholder motifs with documented patterns. Get the Arabic right.

PHASE 03 · Q1 2027

Classroom pilot

Two classrooms, one school. Eight weeks. Track engagement, retention, and bilingual confidence.

PHASE 04 · Q2 2027

Hand off to ADEK

Ship the toolchain ADEK needs to author new chapters: a falconry strand, a date-palm strand, a desert-astronomy strand.

◆ THANK YOU ◆

شكراً لكم

A thread for every Emirati child.

I built *The Pearl and the Loom* to show ADEK what I would do here on day one – and on day three hundred. I would love to do it with you.

SINCERELY,

Bilal Saeed

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LIVE PROTOTYPE

The Pearl and the Loom.html